

Dataset Audit & Preprocessing Strategy

Technical Evaluation of DeepLure Saree Corpus and Kaggle Indian Saree Patterns

1. Dataset Summary

Metric / Attribute	DeepLure Corpus	Kaggle Dataset
Total Images	165 images	1,468 images (410 unique sources)
Labels Provided	None	4 weave categories (Banarasi, Bandhani, Ikat, Pichwai)
Resolution & Framing	226–1512 px; Close-up fabric crops	640×640 (stretched); Product shots
Key Integrity Issues	53 low-texture / near-blank images	48% valid & 43% test images leaked in train split
License & Usage	Proprietary (Evaluation set only)	MIT License (Public Roboflow export)

2. Technical Findings

DeepLure (Evaluation Set)

- Structure:** Close-up weave motifs containing genuine "same design, different colorway" pairs required for evaluation.
- Unlabeled Status:** No design IDs provided. Unsupervised clustering produced unstable counts (108–155 clusters), requiring feature clustering with manual verification.
- Quality Concerns:** 9 near-blank images and 44 plain/low-texture weaves will trigger false matches; these must be quarantined.

Kaggle (Training Signal)

- Split Leakage:** Pre-augmented images were randomly split, causing severe train/val/test leakage (1,468 images boil down to 410 sources).
- Noise & Clutter:** Contains non-fabric objects (jewelry, models, watermarks, collages) and distorted aspect ratios.
- Color Shortcut:** A color histogram alone achieves 46% accuracy on clean splits, demonstrating strong class color bias.

3. Recommended Protocol

Data Cleaning & Preparation

- Deduplicate Kaggle images by perceptual hash and re-split strictly at the cluster level.
- Cluster DeepLure images using edge-normalized feature embeddings, followed by manual review.
- Set up held-out DeepLure evaluation splits with single colorways as queries and remaining colorways as gallery targets.

Training & Evaluation

- Generate synthetic colorways via color jitter, channel permutations, and hue shifts to teach color invariance.
- Mine hard negatives by pairing different weave designs that share identical color palettes.
- Report Recall@1, Recall@5, mAP for retrieval, and ROC-AUC / EER for verification tasks.